ASSIUT UNIVERSITY

Faculty of Engineering — Electrical & Electronics Engineering Department

TECHNICAL REPORT

AC-to-DC Power Supply with Zener Regulation

for a Five-Channel Infrared LED Driver

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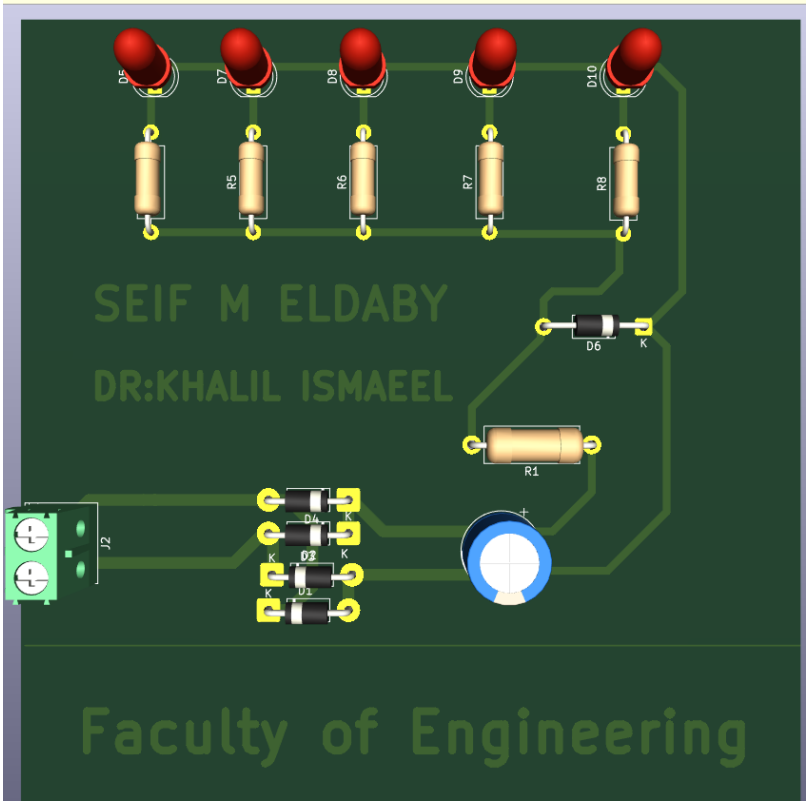


1. Introduction

This report documents the design, analysis, and PCB implementation of an AC-to-DC regulated power supply circuit intended to drive five infrared LEDs . The circuit converts a 50 Hz sinusoidal AC input into a stable regulated DC voltage using a full-wave bridge rectifier, a bulk filter capacitor, a dropping resistor, and a 9.1 V Zener diode regulator. The regulated output is distributed to five identical LED branches, each protected by a dedicated current-limiting resistor.

The project was simulated in LTspice and then implemented as a printed circuit board (PCB). The actual hardware substitutes the schematic's terminal connections with a dedicated 2-pin terminal block (5.08 mm pitch) and uses a 1N4739A (9.1 V, 1 W) Zener diode in place of the BZX84B9V1LY used in simulation.

2. Circuit Description and Block Diagram

2.1 Functional Blocks

The circuit consists of four main stages connected in cascade:

- AC Input Stage — A terminal block accepts the mains-derived sinusoidal input (17 V peak, 50 Hz).
- Full-Wave Bridge Rectifier — Four 1N4007 diodes (D1–D4) form a bridge that rectifies both half-cycles of the AC waveform into a pulsating DC.
- Capacitive Filter — A 2200 µF electrolytic capacitor (C1) smooths the pulsating DC, greatly reducing ripple voltage.
- Zener Shunt Regulator — A 47 Ω series resistor (R1) and a 9.1 V Zener diode (D5, 1N4739A) clamp the output to a stable 9.1 V regardless of moderate load and input variations.
- LED Load Stage — Five parallel branches, each consisting of a 420 Ω resistor in series with LED.

2.2 Component List

Ref.	Description	Value / Part Number	Qty
V1	AC Voltage Source	SINE(0, 17V, 50Hz)	1
D1–D4	Rectifier Diodes	1N4007	4
C1	Filter Capacitor	2200 µF / 50 V Electrolytic	1
R1	Current Limiting / Pre-reg Resistor	47 Ω	1
D5	Zener Voltage Regulator	1N4739A (9.1 V, 1 W)	1
R2–R6	LED Current Limiting Resistors	420 Ω	5
LED1–LED5	Infrared LEDs	White (3.7 v)	5
TB1	Terminal Block	2-Pin, 5.08 mm Pitch	1

3. Design Calculations

3.1 Peak and RMS Input Voltage

The AC source is defined in simulation as SINE(0, 17, 50), meaning:

- Peak voltage: $V_{peak} = 17$ V
- RMS voltage: $V_{rms} = V_{peak} / \sqrt{2} = 17 / 1.4142 \approx 12.02$ V
- Frequency: $f = 50$ Hz $\rightarrow$ Period $T = 20$ ms

3.2 Rectifier Output (Before Filter)

The bridge rectifier drops two diode forward voltages in series during each conduction interval. Using $V_f = 0.7$ V for 1N4007:

- DC average after rectifier (no filter): $V_{dc} = (2 \times V_{peak} / \pi) - 2 \times V_f$
- $V_{dc} = (2 \times 17 / 3.1416) - 1.4 = 10.82 - 1.4 \approx 9.42$ V (unloaded average)
- Peak after rectifier: $V_{peak_rect} = 17 - 1.4 = 15.6$ V

3.3 Filter Capacitor Ripple

With C1 = 2200 µF and the rectifier operating at full-wave frequency (2f = 100 Hz):

- Ripple voltage (approximate): $V_{ripple} = I_{load} / (2 \times f \times C)$
- Estimated total load current (see §3.5): $I_{load} \approx 44$ mA
- $V_{ripple} = 0.044 / (2 \times 50 \times 0.0022) = 0.044 / 0.22 \approx 0.2$ V (peak-to-peak)
- This gives a well-smoothed DC of approximately 15.4 V ± 0.1 V at the capacitor.

3.4 Zener Regulator Stage

The regulated output voltage is set by the Zener diode:

- Zener voltage: $V_Z = 9.1$ V (1N4739A, 1 W power rating)
- Series resistor: R1 = 47 Ω
- Voltage across R1: $V_{R1} = V_{cap} - V_Z = 15.4 - 9.1 = 6.3$ V
- Total current through R1: $I_{R1} = V_{R1} / R1 = 6.3 / 47 \approx 134$ mA
- LED branch current (see §3.5): $I_{LEDs_total} \approx 44$ mA
- Zener current: $I_Z = I_{R1} - I_{LEDs_total} = 134 - 44 = 90$ mA
- Zener power dissipation: $P_Z = V_Z \times I_Z = 9.1 \times 0.090 \approx 0.82$ W (within 1 W rating)
- R1 power dissipation: $P_{R1} = I_{R1}^2 \times R1 = (0.134)^2 \times 47 \approx 0.84$ W (use ≥ 1 W resistor)

3.5 LED Branch Calculations

Each LED branch consists of a 420 Ω resistor in series with a QTLP690C infrared LED. Assuming a typical LED forward voltage $V_{f_LED} = 1.2$ V:

- Voltage across each resistor: $V_{Rn} = V_Z - V_{f_LED} = 9.1 - 1.2 = 7.9$ V
- Current per LED branch: $I_{LED} = V_{Rn} / R_n = 7.9 / 420 \approx 18.8$ mA
- Total LED current (5 branches): $I_{total} = 5 \times 18.8 = 94$ mA
- Power dissipated per resistor: $P_{Rn} = I_{LED}^2 \times 420 = (0.0188)^2 \times 420 \approx 0.148$ W (use ¼ W)
- Power dissipated per LED: $P_{LED} = V_{f_LED} \times I_{LED} = 1.2 \times 0.0188 \approx 22.6$ mW

Note: Recalculating the Zener current with updated load: $I_Z = 134 - 94 = 40$ mA → $P_Z = 9.1 \times 0.040 \approx 0.36$ W. Both scenarios remain within the 1 W Zener rating.

3.6 Summary Table of Key Design Values

Parameter	Calculated Value	Unit
AC Peak Voltage	17	V
AC RMS Voltage	12.02	V
Peak Voltage After Rectifier	15.6	V
Ripple Voltage (p-p)	~0.2	V
Regulated Output Voltage (Zener)	9.1	V
Current Through R1	~134	mA
Zener Current (max case)	~90	mA
Zener Power Dissipation (max)	~0.82	W
Current Per LED Branch	~18.8	mA
Total LED Current	~94	mA
Power Per LED Resistor	~0.148	W

4. Component Selection and Justification

4.1 Rectifier Diodes — 1N4007

The 1N4007 is a general-purpose rectifier diode with a peak inverse voltage (PIV) of 1000 V and average forward current rating of 1 A. The peak inverse voltage seen by each diode in the bridge is equal to the peak supply voltage (17 V), well within the 1000 V rating. The forward current through each diode equals the total load current during conduction, approximately 134 mA, also safely within the 1 A rating.

4.2 Filter Capacitor — 2200 µF

A 2200 µF / 50 V electrolytic capacitor provides sufficient bulk capacitance to reduce the 100 Hz ripple to less than 0.2 V under the expected load. The 50 V voltage rating provides more than three times the expected peak voltage (15.6 V), ensuring long-term reliability. Larger capacitance could reduce ripple further but would increase inrush current and PCB footprint.

4.3 Zener Diode — 1N4739A (9.1 V, 1 W)

The 1N4739A is a through-hole 1 W Zener diode in a DO-41 package with a regulation voltage of 9.1 V. It replaces the SMD BZX84B9V1LY used in simulation, offering the same regulation point with higher power dissipation capability (1 W vs. 300 mW). This is important because the Zener may need to sink up to ~90 mA under light load, dissipating up to 0.82 W, which exceeds the capability of the 300 mW SMD variant.

4.4 LED Current-Limiting Resistors — 420 Ω

A resistance of 420 Ω limits the LED branch current to approximately 18.8 mA, within the absolute maximum forward current of 30 mA. This provides adequate IR radiant intensity while keeping power dissipation per resistor below 0.15 W, suitable for standard 1/4 W carbon film resistors.

4.5 Terminal Block — 2-Pin, 5.08 mm Pitch

The 5.08 mm pitch terminal block provides a robust, tool-accessible screw connection for the AC input wires. It replaces any loose wire connections in the prototype and is rated for currents well above the circuit's demand, improving safety and reliability in the final PCB assembly.

5. Methodology

5.1 Simulation in LTspice

The circuit was first drawn and simulated in LTspice using a transient analysis command (.tran 0.5) to observe 500 ms of circuit behaviour. The simulation confirmed correct rectification, filtering, and voltage regulation at 9.1 V, as well as individual LED branch currents within specification.

Simulation Results

The following image placeholders should be replaced with screenshots exported from LTspice.

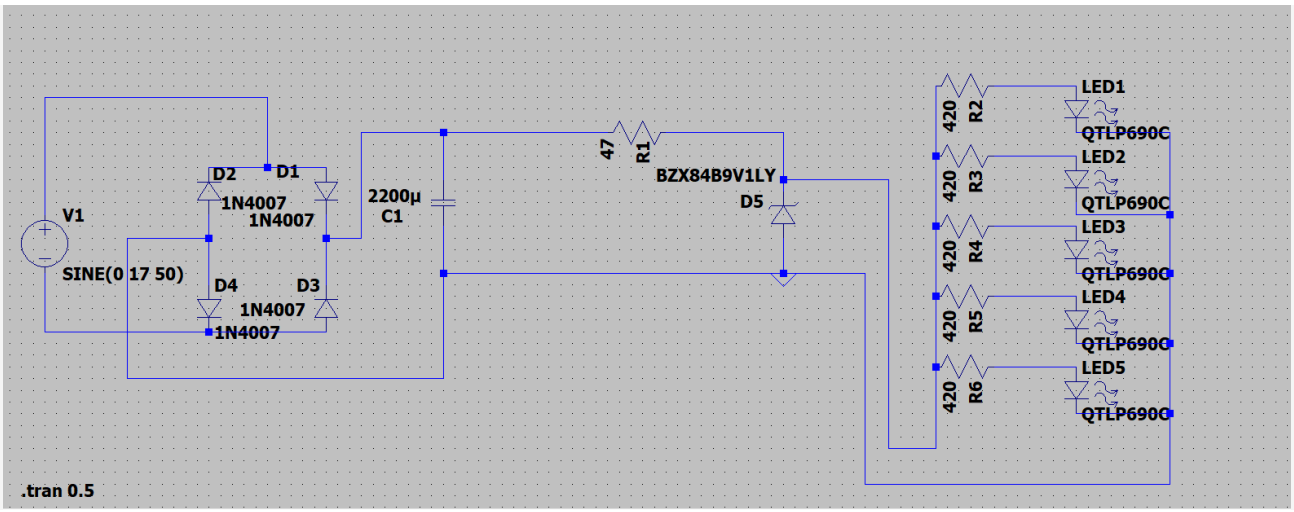


Figure 1 — LTspice Schematic Screenshot

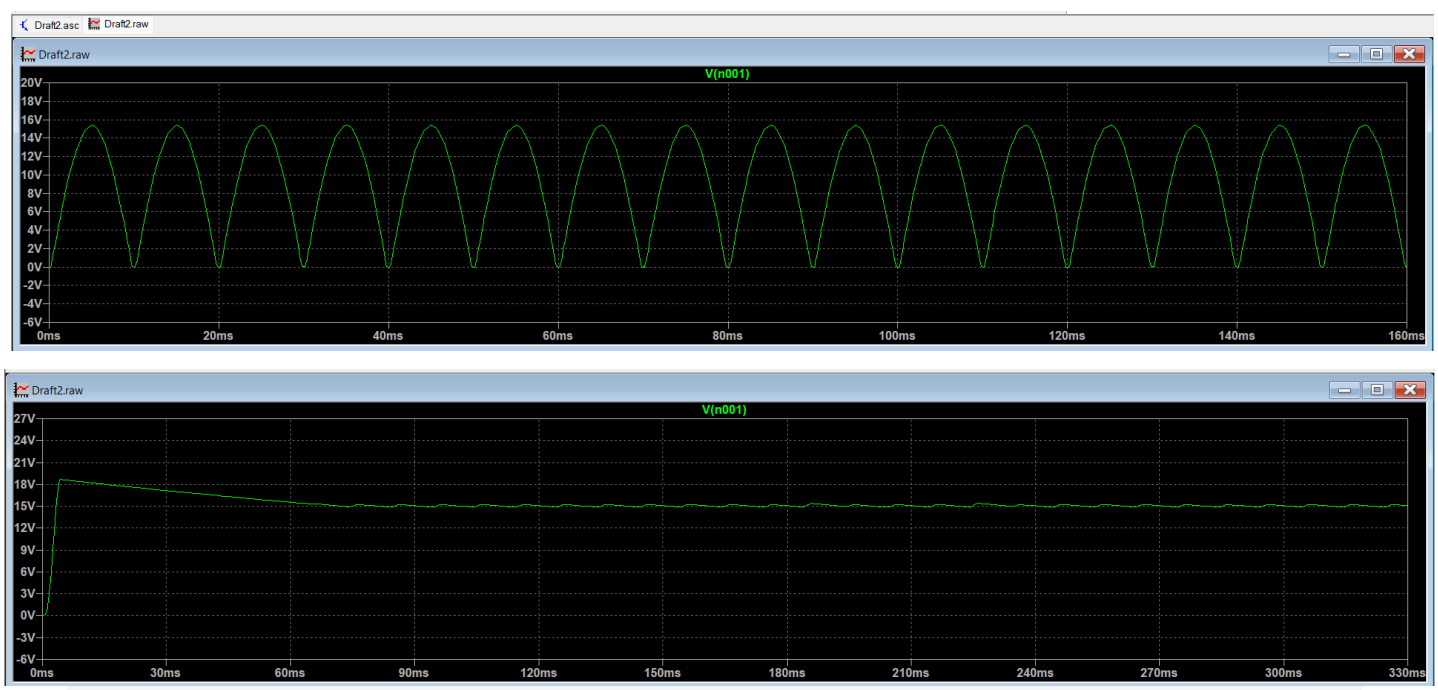


Figure 2 — Output Waveform: Rectified and Filtered DC Voltage (V_cap)

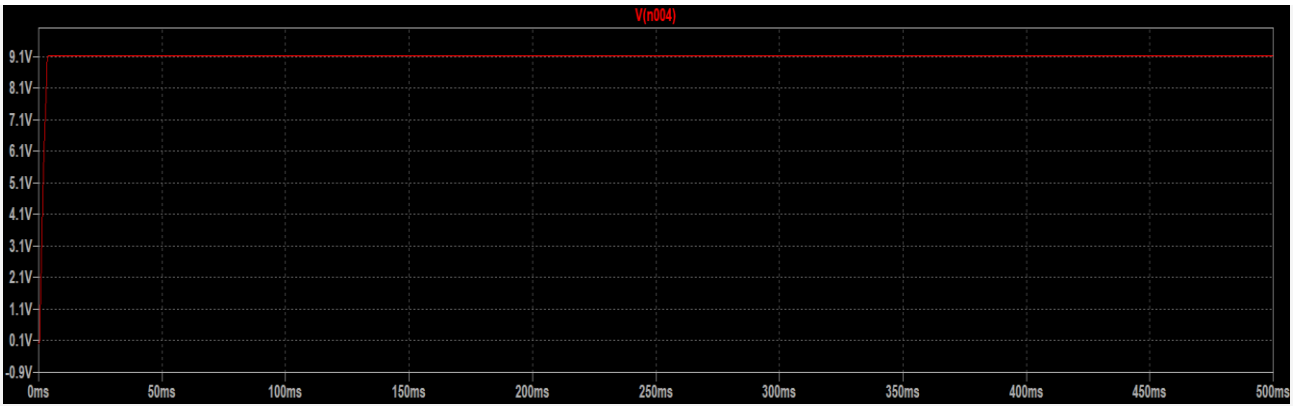


Figure 3 — Output Waveform: Regulated Zener Output (9.1 V)

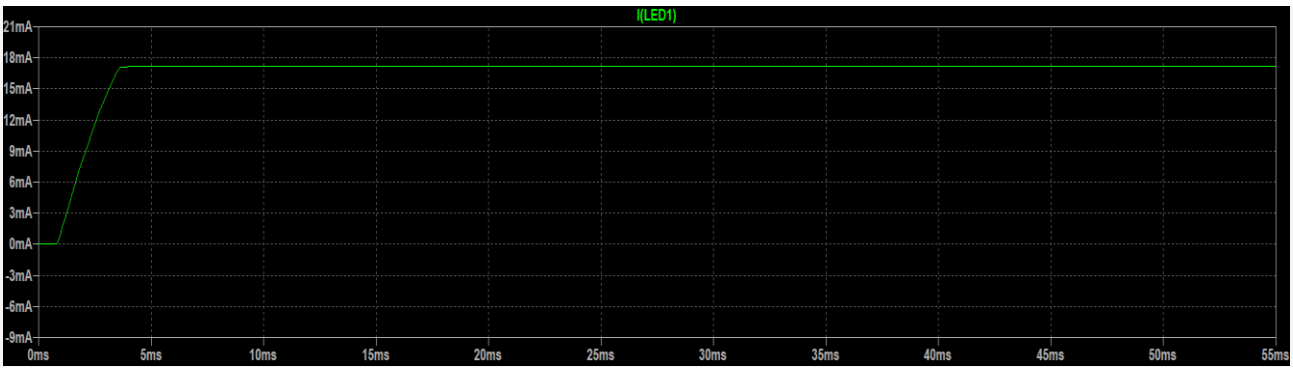


Figure 4 — LED Branch Currents (I_LED1 through I_LED5)

5.2 PCB Design

Following successful simulation, the schematic was transferred to a PCB layout tool. The following design decisions were applied:

- AC input traces routed with adequate creepage and clearance distance from DC traces (minimum 2 mm for 50 V working voltage, per IEC 60950).
- The Zener diode and series resistor R1 placed close together to minimise high-current trace length.
- LED driver resistors (R2–R6) arranged in a column mirroring the LED positions for a clean, symmetrical layout.
- 2-pin terminal block mounted at the board edge for direct cable access.
- All through-hole components on the top layer; solder joints on the bottom layer.

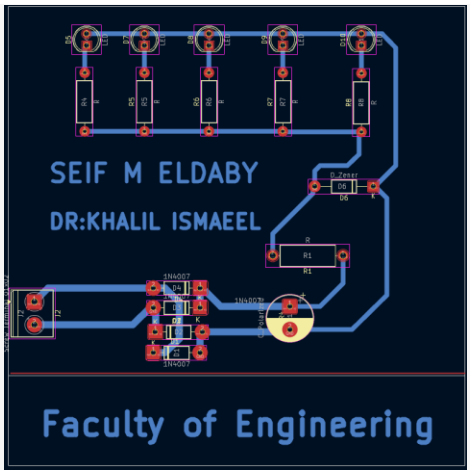


Figure 5 — PCB Top Layer

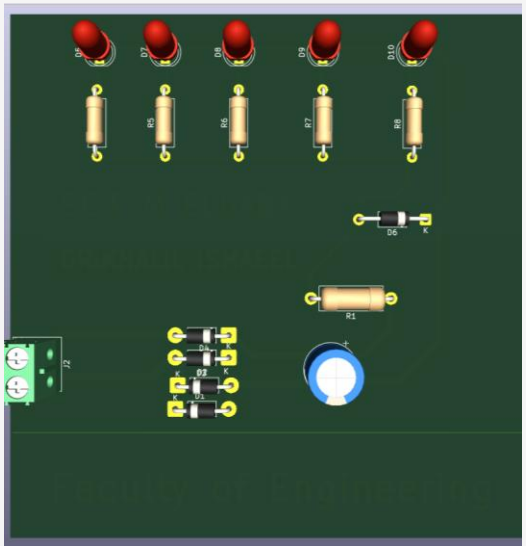


Figure 6 — PCB 3D Render (Top View)

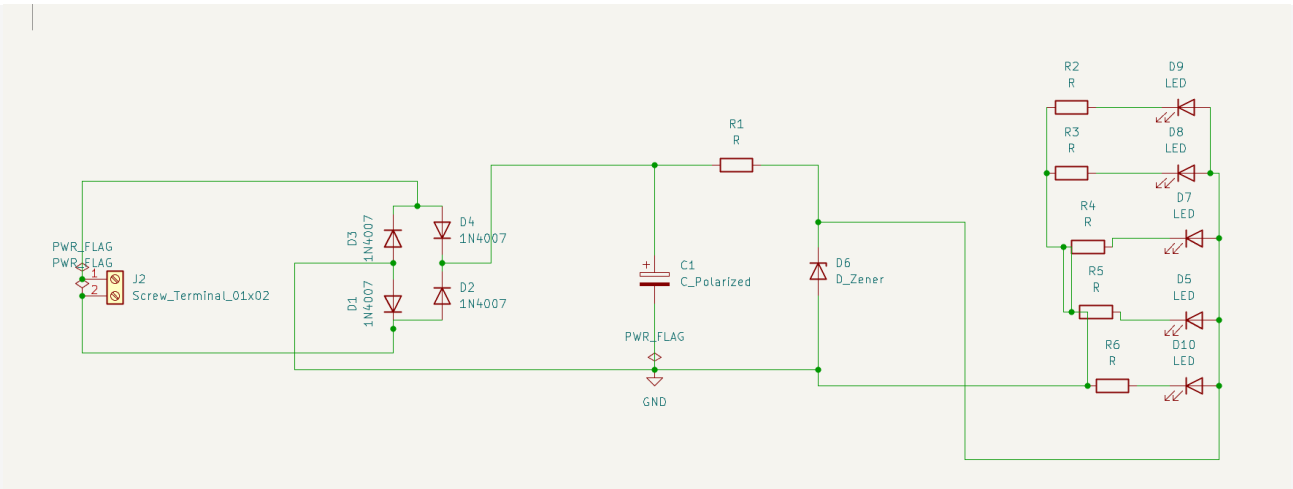


Figure 7 — schematic on kicad

6. Results and Discussion

The simulation results confirmed that the full-wave bridge rectifier produces a pulsating DC with a peak of 15.6 V. After filtering with the 2200 μF capacitor, the ripple was reduced to approximately 0.2 V peak-to-peak, yielding a stable pre-regulation supply of ~15.4 V.

The Zener shunt regulator successfully clamped the output at 9.1 V. The series resistor R1 = 47 Ω carried approximately 134 mA, shared between the Zener diode and the LED load. Each of the five LED branches drew approximately 18.8 mA, producing sufficient infrared radiance from the QTLP690C LEDs while remaining well below the absolute maximum forward current.

The substitution of the SMD Zener (BZX84B9V1LY) with the 1 W through-hole 1N4739A in the PCB implementation ensures adequate thermal margin. Power dissipation calculations confirmed that the Zener operates within its 1 W rated maximum under all expected load conditions.

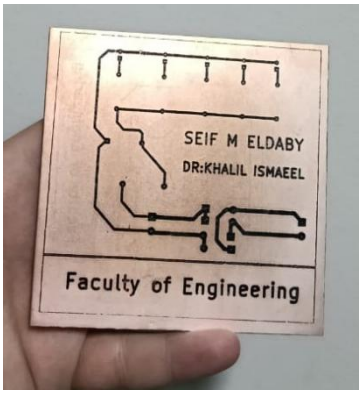
7. Conclusion

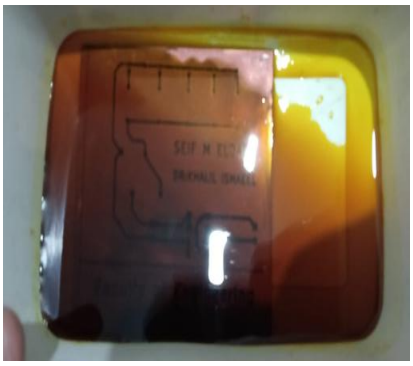
A compact AC-to-DC regulated LED driver circuit has been successfully designed, simulated, and implemented on a printed circuit board. The design achieves a stable 9.1 V regulated output from a 12 V RMS / 17 V peak AC source and distributes approximately 18.8 mA to each of LEDs. All components were selected with adequate voltage, current, and power margins. The PCB layout adheres to good engineering practices regarding trace routing, component placement, and creepage/clearance requirements.

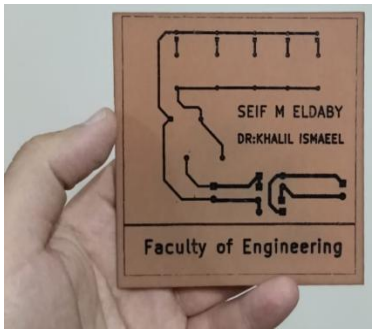
Manufacturing Process Steps

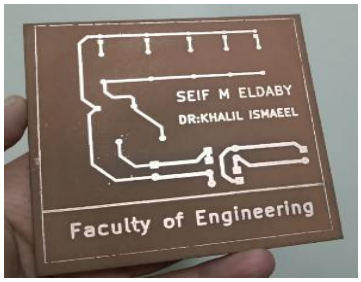
In this section, the manual fabrication process of the PCB is detailed through the following stages:

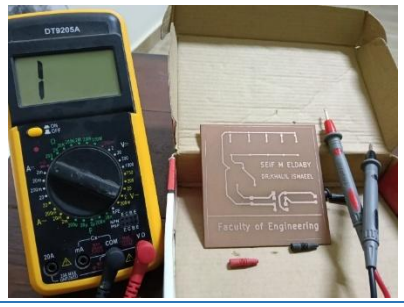
- **1. Toner Transfer:** The circuit layout was printed using a laser printer on glossy paper and then transferred onto the copper-clad board using heat (ironing method).
- **2. Chemical Etching:** The board was immersed in an etching solution (Ferric Chloride) to remove the unwanted copper, leaving only the protected traces.
- **3. Cleaning and Inspection:** After etching, the toner was removed using a solvent (like Thinner or Acetone) to reveal the clean copper tracks. A visual inspection was performed to ensure no shorts or breaks.
- **4. Continuity Testing:** Using a **Digital Multimeter**, a continuity test was conducted to verify the electrical integrity of all traces before proceeding to the drilling and soldering phases.
- **5. Component Preparation:** The electronic components (LEDs, capacitors, and terminal blocks) were prepared and organized for the final assembly.
- **6. Drilling:** Precision holes were drilled for the through-hole components (THD) using a mini-drill press. Drill bit sizes were selected based on component lead diameters (typically **0.8mm** to **1mm**).
- **7. Component Assembly & Soldering:** Components were placed according to the silk-screen layout. Soldering was performed using a temperature-controlled soldering iron and high-quality lead-tin wire, ensuring "concave" solder joints for maximum mechanical and electrical reliability.
- **8. Final Testing & Calibration:** After assembly, the circuit was connected to the AC source through a safety isolation transformer (or series lamp) for initial power-up. Output voltage and current were measured using the **multimeter** to ensure they met the LED driver specifications.
- **9. Enclosure & Final Inspection:** A final visual check was done to ensure no cold solder joints or bridge shorts existed. The board was then cleaned with Iso-Propyl Alcohol (IPA) to remove flux residues, giving it a professional finish.

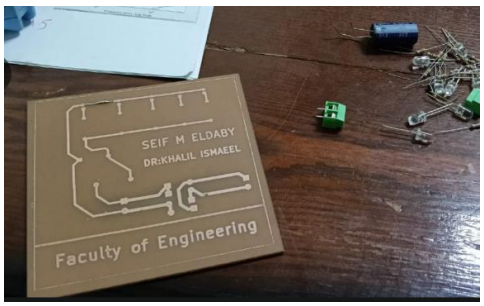


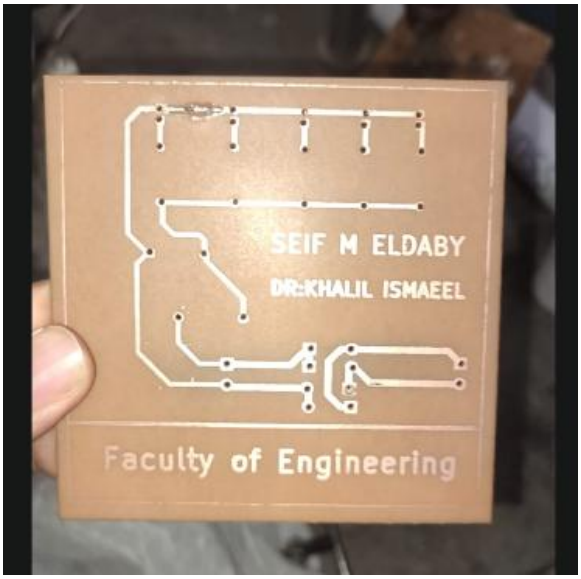
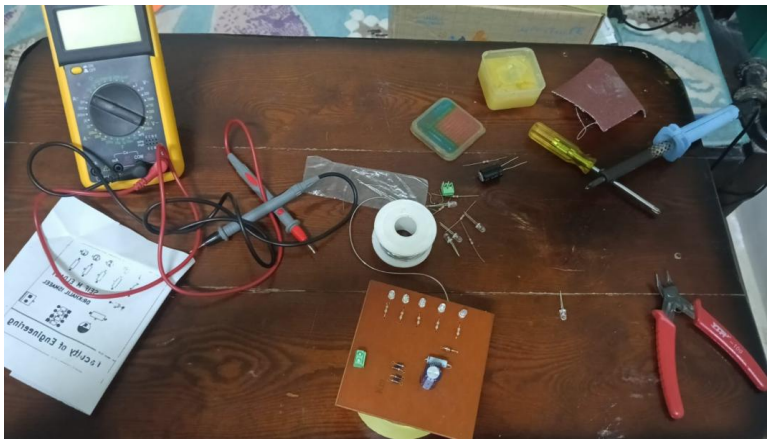
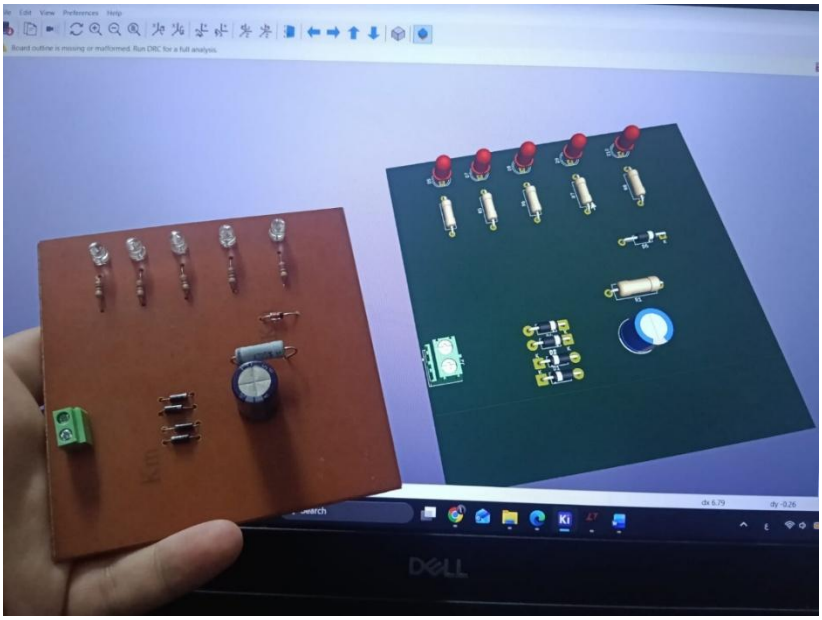
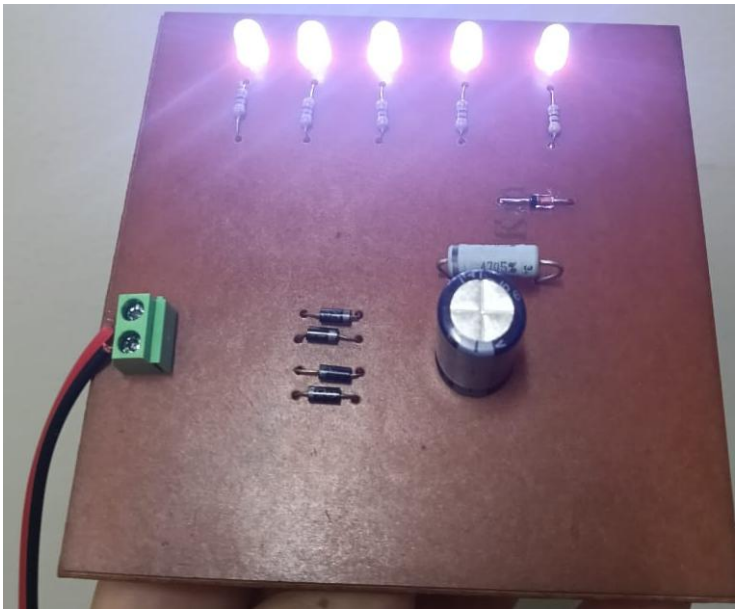








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